Solstice Task 2 - PART A

Mathematics Club

Instructions

- Provide descriptive answers for all questions. Explain your thought process well.
- You are not expected to understand anything in first read. Let it sink in. Ask doubts, discussions highly encouraged.
- It is suggested and expected to not use LLMs (ChatGPT, Gemini, etc.) for thinking purposes. Feel free to use it for understanding purpose but not blindly. If the question asks you to derive/prove something, have fun struggling solving the problem and feeling the joy as you get closer to the answer. If you find yourself utterly stuck, do ask for hints on group or DM! One hour left before submission, hints did not work, go get some rescue from GPT. Although be mindful to not trust blindly and note that you are much smarter in thinking than GPT.

All the best and have fun!

§1 We love integration, don't we?

Let $p(x|w)$ and $\varphi(w)$ be a statistical model and a prior, where $x \in \mathbb{R}^N, w \in W \subset \mathbb{R}^d$. The Hamiltonian function $H(w)$ is defined by

$$H(w) = -\sum_{i=1}^{n} \log p(X_i|w) - \log \varphi(w).$$

Then the posterior distribution is represented by

$$\begin{aligned} p(w) &= \frac{1}{Z_n} \varphi(w) \prod_{i=1}^{n} p(X_i|w) \\ &= \frac{1}{Z_n} \exp(-H(w)), \end{aligned}$$

where Z_n is a partition function or the marginal likelihood,

$$Z_n = \int \exp(-H(w)) dw.$$

The probability density function $p(w)$, which is equal to the posterior distribution in Bayesian statistics, is called the equilibrium state of the Hamiltonian $H(w)$. Our purpose in this section is to generate $\{w_k\}_{k=1}^{K}$ such that, for an arbitrary function $f(w)$,

$$\int f(w)p(w)dw \approx \frac{1}{K} \sum_{k=1}^{K} f(w_k) \tag{1}$$

when $K \to \infty$. Think of it like, because it is not possible to directly calculate the integral on left, we take many values of the function f and average them, so that it approximates the integral on LHS.

In most cases in statistical applications, n is large, hence the parameter set

$$\{w \in W; p(w) > \epsilon\}$$

for some $\epsilon > 0$ is a very narrow subset of W, resulting that Riemann sum of the integral on the parameter space does not give effective approximation. Or in other words the distribution $p(w)$ is very low mostly with some huge spikes in middle (maybe) and so taking average over all $f(w)$ without weighing by $p(w)$ is a bad approximation.

Remark (1) (Curse of dimensionality) If $d = 1$, then the integral can be approximated by Riemann sum,

$$\int_0^1 f(w)p(w)dw \approx \frac{1}{K}\sum_{k=1}^{K} f(k/K)p(k/K),$$

which is more accurate than MCMC. However, if $d = 2, 3, 4, \cdots$, then

Why is K^d a problem?

The problem is **Exponential Explosion**. The number doesn't just get "big", it gets "physically impossible" very quickly.

Let's say we want a decent approximation, so we cut each dimension into just 10 slices ($K = 10$).

- **1 Dimension ($d = 1$):**
  You have a line. You need $10^1 = 10$ points.
  Computer time: Nanoseconds. Easy.
- **2 Dimensions ($d = 2$):**
  You have a square. You need $10^2 = 100$ points.
  Computer time: Microseconds. Easy.
- **3 Dimensions ($d = 3$):**
  You have a cube. You need $10^3 = 1,000$ points.
  Computer time: Still easy.
- **20 Dimensions ($d = 20$):**
  (Note: In AI and Statistics, 20 parameters is considered a *tiny* model).
  You need 10^{20} points.
  10^{20} **is 100 Quintillion.**
  If you had a supercomputer that could calculate **1 trillion points per second**, it would take **3,170 years** to finish this calculation.
- **100 Dimensions ($d = 100$):**
  You need 10^{100} points.
  There are only about 10^{80} atoms in the entire observable universe.
  Even if every atom in the universe were a supercomputer working for the entire age of the universe, you wouldn't make a dent in this calculation.

Hence to fix both these problems (most of $w \in W$ has low p, curse of dimensionality), we use MCMC- Monte Carlo Markov Chain. This ignores the empty space. It uses a "smart walker" that only steps where the probability is high. It doesn't care about d nearly as much.

In the Markov chain Monte Carlo (MCMC) method, a sequence $\{w_1, w_2, w_3, \dots\}$ is generated by a conditional probability $p(w_{k+1}|w_k)$ iteratively. It is known that (1) and (2) below are sufficient conditions for (1) to always hold, i.e. we will always get approximation of the expectation.
(1) **Detailed Balance Condition**. For arbitrary parameters $w_a, w_b \in W$,

$$p(w_b|w_a)p(w_a) = p(w_a|w_b)p(w_b).$$

(2) **Irreducible Condition**. For an arbitrary $w \in W$, the probability that a parameter of $\{w_k\}$ is contained in the neighborhood of w is not equal to zero.

(Note that the detailed balance condition is not necessary for eq.(1), however, there are several MCMC algorithms which satisfy the detailed balance condition.)
Intution:
(2) tries to make sure that theoretically, it is possible to sample from the entire (valid $p(w) \neq 0$) space W. (1) Ensures that the probability of moving from w_a to w_b is same as that of w_b to w_a so that the samples are not heavily populated in one region.

Firstly, we study Metropolis method.

§1 Metropolis Method

1. Initialize $w(1)$ and $t = 1$.
2. A candidate w' is generated by $r(w'|w(t))$.
3. By using $\Delta H \equiv H(w') - H(w(t))$, the probability $P = \min\{1, \exp(-\Delta H)\}$ is determined. Then set $w(t+1) = w'$ with probability P, or $w(t+1) = w(t)$ with probability $1 - P$. (So if according to Hamiltonian w' is more favored, P=1 i.e. we always move to w', otherwise we move to w' with probability $\exp(-\Delta H)$, i.e. sometimes we will and sometimes we won't).
4. $t := t + 1$, and return to (2).

We can prove that this procedure satisfies the detailed balance condition.

Distribution r

- **What is it?** $r(w'|w(t))$ is a probability distribution (often a Gaussian) centered at your current location that defines where you *might* jump next.
- **Symmetry:** The two MCMC conditions requires that r is fair: $r(A|B) = r(B|A)$. The chance of suggesting a jump from $A \to B$ must equal the chance of suggesting $B \to A$.
- **Distribution vs. Sample:** In step (2) of the algorithm, r is the distribution, but w' is a specific **realization** (sample) drawn from it. We use this single concrete point w' to calculate ΔH.

Reflect: There are various limitations of Metropolis method. What do you think could they possibly be? Read up from given resources after thinking.

§2 Gibbs Sampler

We want to sample from a complex joint distribution $p(w_1, w_2)$. We assume we cannot sample from this directly, but we *can* sample from the "conditional" distributions:

- $p(w_1|w_2)$: "If I know w_2, what is the probability of w_1?"
- $p(w_2|w_1)$: "If I know w_1, what is the probability of w_2?"

Algorithm

1. **Initialize:** Start with an initial guess $w(1) = (w_1(1), w_2(1))$ and set time $t = 1$.
2. **Choose Update Order:** With probability 1/2, choose either path (A) or (B):
   - **(Path A):**
     - First, update w_2 based on the *old* w_1: Sample $w_2' \sim p(w_2|w_1(t))$.
     - Then, update w_1 based on the *new* w_2': Sample $w_1' \sim p(w_1|w_2')$.
   - **(Path B):**
     - First, update w_1 based on the *old* w_2: Sample $w_1' \sim p(w_1|w_2(t))$.
     - Then, update w_2 based on the *new* w_1': Sample $w_2' \sim p(w_2|w_1')$.
3. **Update & Repeat:** Set the new state $w(t+1) = (w_1', w_2')$, increment t, and repeat.

Example

Let's visualize a 2D Normal distribution where x and y are highly correlated (a narrow, diagonal valley).

$$p(x, y) \propto \exp\left(-10(x-y)^2 - 0.1(x+y)^2\right) \tag{1}$$

- **Metropolis:** Would try to jump in a random circle. Most jumps land on the steep walls of the valley and are rejected.
- **Gibbs:** Moves strictly North/South or East/West.

Iteration 1

Start: We are at $(0, 0)$.

Step 2 (Path A chosen):

- **Update y given $x = 0$:** The conditional distribution $p(y|x = 0)$ is a Gaussian centered near 0.
  We sample: $y_{\text{new}} = 1.0$.
  Current Position: $(0, 1.0)$.
- **Update x given $y = 1.0$:** Now we look at $p(x|y = 1.0)$. Because they are correlated, the mean of x shifts towards y.
  We sample: $x_{\text{new}} = 0.8$.
  Current Position: $(0.8, 1.0)$.

Iteration 2

Step 2 (Path B chosen):

- **Update x given $y = 1.0$:** We sample a new x again.
  Sample $x_{\text{new}} = 1.2$.
  Position: $(1.2, 1.0)$.
- **Update y given $x = 1.2$:**
  Sample $y_{\text{new}} = 1.3$.
  Position: $(1.2, 1.3)$.

Result

The sampler "zigzags" its way up the diagonal.

$$(0,0) \to (0,1) \to (0.8,1) \to (1.2,1) \to (1.2,1.3)\ldots$$

Notice it never moves diagonally in one step. It creates a "staircase" pattern. This is efficient because we never get rejected, but inefficient because we take many small steps to cover a large distance. To overcome this, we have momentum based method- HMC.

§2.1 Why Gibbs Sampling Works (and when it fails)

To understand Gibbs sampling deeply, we must view it through the lens of the more general **Metropolis-Hastings** framework and analyze its behavior on a **Gaussian** surface.

¶ Task 1: Gibbs as a Special Case of Metropolis-Hastings **1. The General Metropolis-Hastings Rule**
In the general Metropolis-Hastings algorithm, to sample from a target distribution $p(w)$, we propose a move from current state w to a new state w' using a proposal distribution $q(w'|w)$. The move is accepted with probability $\alpha(w, w')$:

$$\alpha(w, w') = \min\left(1, \frac{p(w')}{p(w)} \cdot \frac{q(w|w')}{q(w'|w)}\right)$$

- $p(w)$ and $p(w')$ are the probabilities of the states under the target distribution.
- $q(w'|w)$ is the probability of proposing w' given we are at w.
- $q(w|w')$ is the probability of proposing the reverse move (going back to w) given we are at w'.

2. The Gibbs Proposal Definition
In a Gibbs update step for the i-th component, we hold all other variables w_{-i} fixed.

- **Current State:** $w = (w_i, w_{-i})$
- **Proposed State:** $w' = (w'_i, w_{-i})$ (Note: w_{-i} is identical in both).

The Gibbs algorithm proposes the new value w'_i by sampling directly from the conditional distribution of the target. Therefore, the proposal distribution q is defined as:

$$q(w'|w) = p(w'_i|w_{-i})$$

Conversely, the reverse proposal (going back to w_i from w'_i) is:

$$q(w|w') = p(w_i|w_{-i}) \quad (\text{since } w'_{-i} = w_{-i})$$

3. Proof Steps
Goal: Prove that plugging these definitions into the acceptance ratio formula results in $\alpha = 1$.
Step 1: Factorize the Target Distribution
Using the chain rule of probability, $p(a, b) = p(a|b)p(b)$, express the joint probability $p(w)$ in terms of the component w_i and the rest w_{-i}.

$$p(w) = p(w_i, w_{-i}) = p(w_i|w_{-i}) \cdot p(w_{-i})$$

Write the equivalent expression for $p(w')$.

Step 2: Formulate the Ratio

Substitute your expressions from Step 1 and the definitions of $q(w'|w)$ and $q(w|w')$ into the fraction:

$$\text{Ratio} = \frac{p(w')}{p(w)} \cdot \frac{q(w|w')}{q(w'|w)}$$

Step 3: Show Cancellation

1. Substitute the factorized $p(w')$ and $p(w)$ into the ratio.
2. Substitute the conditional proposals $p(w_i|w_{-i})$ and $p(w_i'|w_{-i})$.
3. Show that all terms cancel out perfectly, proving that Ratio $= 1$.

Conclusion: Since the ratio is 1, $\alpha = \min(1, 1) = 1$. This proves that Gibbs sampling updates are **always accepted**.

¶ Task 2: Mixing Time and Condition Number

Background: The Bivariate Gaussian

Let's make the target distribution concrete. Consider a 2D Gaussian centered at 0 with unit variance and correlation ρ:

$$\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$$

The correlation coefficient $\rho \in (-1, 1)$ controls how "stretched" the distribution is.

- $\rho = 0$: A perfect circle (variables are independent).
- $\rho \to 1$: A infinitely thin diagonal line (variables are identical).

Background: Mixing Time

Mixing Time is a measure of "how long it takes for the sampler to converge." Imagine dropping a drop of ink (the sampler) into a glass of water (the probability space).

- **Fast Mixing:** The ink spreads to the whole glass instantly. (Sampler jumps across the space easily).
- **Slow Mixing:** The ink stays in a tight blob. (Sampler gets stuck in one region).

Background: Conditional Variance Formula

For a Bivariate Gaussian, if we know w_2, the conditional distribution of w_1 is also a Gaussian:

$$p(w_1|w_2) = \mathcal{N}(\mu_{1|2}, \sigma^2_{1|2})$$

The formula for the **conditional variance** (how much we wiggle) is:

$$\sigma^2_{1|2} = \sigma^2_1(1 - \rho^2)$$

(Since we assumed unit variance, $\sigma^2_1 = 1$).

Proof Steps

Goal: Show that high correlation ($\rho \approx 1$) breaks the sampler.

Step 1: The Limit of Variance

Using the conditional variance formula above:

$$\text{Variance of Update} = 1 - \rho^2$$

Calculate the limit of this variance as $\rho \to 1$.

Step 2: Interpret the Step Size

In Gibbs sampling, the "step size" is determined by this variance. We sample w_1' from a window of width $\sqrt{1-\rho^2}$.

- If the variance is near 0, how far can the sampler move from its current position?
- Explain why this leads to "slow mixing" (taking millions of steps to cross the distribution).

Step 3: Connect to Condition Number (Advanced)

The **Condition Number** κ of a matrix is the ratio of its largest to smallest eigenvalue: $\kappa = \lambda_{\max}/\lambda_{\min}$. The eigenvalues of our correlation matrix Σ are $(1+\rho)$ and $(1-\rho)$.

$$\kappa(\Sigma) = \frac{1+\rho}{1-\rho}$$

- As $\rho \to 1$, what happens to κ?
- *Conclusion:* A high condition number (ill-conditioned problem) corresponds to a "narrow valley" geometry, which forces Gibbs sampling into tiny, inefficient steps.

NOTE: BELOW SECTION IS OPTIONAL

§3 Hamiltonian Monte Carlo

Hamiltonian Monte Carlo (HMC) augments the target distribution $p(w)$ by introducing an auxiliary momentum variable $v \in \mathbb{R}^d$. The joint system is described by the **Hamiltonian** $\mathcal{H}(w,v)$, which represents the total energy of the system:

$$\mathcal{H}(w,v) = H(w) + K(v) = H(w) + \frac{1}{2}\|v\|^2 \tag{2}$$

where:

- $H(w) = -\log p(w)$ is the **Potential Energy** (derived from the target distribution).
- $K(v) = \frac{1}{2}v^T v$ is the **Kinetic Energy** (typically chosen as a standard Gaussian).

The evolution of the system over a fictitious time τ is governed by Hamilton's equations of motion:

$$\frac{dw}{d\tau} = \frac{\partial \mathcal{H}}{\partial v} = v \tag{3}$$

$$\frac{dv}{d\tau} = -\frac{\partial \mathcal{H}}{\partial w} = -\nabla H(w) \tag{4}$$

These equations describe a conservative system where $\frac{d\mathcal{H}}{d\tau} = 0$. The HMC algorithm alternates between two steps:

1. **Momentum Resampling:** Sample a new momentum $v \sim \mathcal{N}(0, I)$. This changes the energy level (switching "orbits").
2. **Hamiltonian Dynamics:** Simulate the system for time τ to propose a new state (w', v'). Due to energy conservation, $\mathcal{H}(w', v') \approx \mathcal{H}(w, v)$, leading to a high acceptance probability.

Example:

Consider a target distribution that is a standard 1D Gaussian, $w \sim \mathcal{N}(0, 1)$. The potential energy is:

$$H(w) = \frac{1}{2}w^2 \quad \Longrightarrow \quad \nabla H(w) = w \tag{5}$$

Question:Substitute this into Hamilton's equations, what equations do you yield? Have you seen them?
Hint: The analytical solution for the state at time τ, given initial state (w_0, v_0), is a rotation in phase space:

$$w(\tau) = w_0 \cos(\tau) + v_0 \sin(\tau) \tag{6}$$
$$v(\tau) = v_0 \cos(\tau) - w_0 \sin(\tau) \tag{7}$$

Numerical Illustration

Let the current parameter be $w_0 = 1$. We sample a random momentum $v_0 = -1$. If we integrate for time $\tau = \pi$ (a half-period):

$$w(\pi) = (1)\cos(\pi) + (-1)\sin(\pi) = -1$$
$$v(\pi) = (-1)\cos(\pi) - (1)\sin(\pi) = 1$$

The sampler moves from $w = 1$ to $w = -1$ in a single deterministic step. The total energy remains perfectly conserved:

$$\mathcal{H}_{\text{initial}} = \frac{1}{2}(1)^2 + \frac{1}{2}(-1)^2 = 1 \quad \rightarrow \quad \mathcal{H}_{\text{final}} = \frac{1}{2}(-1)^2 + \frac{1}{2}(1)^2 = 1$$

Since $\Delta\mathcal{H} = 0$, the acceptance probability is $\min(1, \exp(0)) = 1$.

Random Momentum and Hyperparameters

Does random v worsen the state? Sampling a random v might propel the parameter w into a low-probability region (high potential energy $H(w)$).

This does not degrade the sampler performance because of **Energy Conservation**. If the dynamics drive w up a steep potential slope (increasing $H(w)$), the kinetic energy $K(v)$ must decrease by an equal amount to keep $\mathcal{H}$ constant.

- The proposal (w', v') will have nearly the same total energy as (w, v).
- The acceptance probability depends on $\Delta\mathcal{H} \approx 0$.

Therefore, even if w' is in a "worse" location (lower probability density), the move is likely to be accepted because the momentum v' has compensated for it. This allows the sampler to efficiently explore the tails of the distribution.

The Integration Time τ

The parameter τ determines how long we run the physical simulation.

- **Small τ:** The system resembles a random walk (inefficient exploration).
- **Large τ:** The trajectory may form a closed loop (U-turn), returning to the starting point and wasting computation.
- **Optimal τ:** We ideally want to integrate until the path makes a U-turn, maximizing the distance from the start point.

Example:
Let's go back to our harmonic oscillator (a marble rolling in a bowl).
Start: Release the marble at the top of the left rim ($w = -5$).
Physics: It rolls down to the bottom (0) and up the other side.
Optimal Point: It reaches the top of the right rim ($w = +5$). This is the furthest possible point from your start. You should stop here!
If you let the simulation run longer (Large τ), gravity pulls it back down, and it rolls all the way back to the left rim ($w = -5$).If you stop the simulation after the U-turn, you tell your algorithm: "My new random sample is... the exact same number I started with." That is inefficient.

Note that according to our distribution (Normal gaussian), -5 and 5 had equal probabilites, here it was due to τ that sampling was affected.

In modern implementations, τ is not set manually. Algorithms like the **No-U-Turn Sampler (NUTS)** automatically determine the optimal integration time adaptively during each step, eliminating the need to tune this hyperparameter manually.

How do we do run the physics simulations analytically? **Leap Frog method**

To truly understand why HMC works, we need to look under the hood of the **Leapfrog Integrator**. It is not just a random choice; it has two special geometric properties: **Volume Preservation** and **Symplectic Error Bound**.

§3.1 Volume Preservation (Liouville's Theorem)

The Leapfrog Map The Leapfrog method updates the state (w, v) to (w^*, v^*) over a small time step ϵ by splitting the physics into three sub-steps:

1. **Half-Kick (T_1):** Update momentum by half a step using the gradient at current position.

$$v_{1/2} = v - \frac{\epsilon}{2}\nabla H(w)$$

2. **Drift (T_2):** Update position by a full step using the new half-momentum.

$$w^* = w + \epsilon v_{1/2}$$

3. **Half-Kick (T_3):** Update momentum by the remaining half step using the gradient at the new position.

$$v^* = v_{1/2} - \frac{\epsilon}{2}\nabla H(w^*)$$

We can view the full update Φ as a composition of these functions: $\Phi = T_3 \circ T_2 \circ T_1$.

Background: The Jacobian To check if a transformation compresses or expands space (volume), we look at its **Jacobian Matrix** J. For a map $F : \mathbb{R}^k \to \mathbb{R}^k$, the Jacobian is the matrix of all partial derivatives:

$$J_{ij} = \frac{\partial F_i}{\partial x_j}$$

The **Determinant** of this matrix, $\det(J)$, tells us the volume scaling factor.

- If $|\det(J)| = 1$, the map is **Volume Preserving**.

- If $|\det(J)| < 1$, the map shrinks space (like a sink).
- If $|\det(J)| > 1$, the map expands space (like a source).

Proof Steps
Goal: Prove that $|\det(J_\Phi)| = 1$.
1. Chain Rule
Using the chain rule for determinants, we know that:

$$\det(J_\Phi) = \det(J_{T_3}) \cdot \det(J_{T_2}) \cdot \det(J_{T_1})$$

We just need to prove the determinant of each sub-step is 1.
2. Analyze T_1 (The Shear Map)
Let's look at the first half-kick $T_1(w, v) \to (w, v_{1/2})$.

$$\begin{cases} w_{new} = w_{old} \\ v_{new} = v_{old} - \dfrac{\epsilon}{2}\nabla H(w_{old}) \end{cases}$$

The Jacobian matrix has the block structure:

$$J_{T_1} = \begin{pmatrix} \dfrac{\partial w_{new}}{\partial w_{old}} & \dfrac{\partial w_{new}}{\partial v_{old}} \\ \dfrac{\partial v_{new}}{\partial w_{old}} & \dfrac{\partial v_{new}}{\partial v_{old}} \end{pmatrix} = \begin{pmatrix} I & 0 \\ -\dfrac{\epsilon}{2}\mathbf{H}_H(w) & I \end{pmatrix}$$

where $\mathbf{H}_H(w)$ is the Hessian of H. Since this is a **lower triangular matrix**, its determinant is just the product of the diagonal entries.

$$\det(J_{T_1}) = 1 \cdot 1 = 1$$

3. Analyze T_2 and T_3
Construct the Jacobian matrices for the Drift step (T_2) and the second Half-Kick (T_3).

- For T_2: Show it is an *upper* triangular matrix. What is its determinant?
- For T_3: Show it is a *lower* triangular matrix (similar to T_1).

4. Conclusion
Combine the determinants to show the total volume scaling is 1.

Reflection Question In the Metropolis-Hastings acceptance ratio:

$$\alpha = \min\left(1, \frac{p(w')}{p(w)} \cdot \frac{q(w|w')}{q(w'|w)} \cdot |\det(J)|\right)$$

Usually, when we change variables, we must include the Jacobian term. Because you just proved $|\det(J)| = 1$, what does this simplify to? (Hint: This is why HMC doesn't need complex Jacobian corrections).

§3.2 Geometric Integration Error

Background: Order of Integrators Numerical integrators are essentially approximations of Taylor Series.

- An integrator is of **Order** p if its **Local Error** (error after one single step of size ϵ) is $O(\epsilon^{p+1})$.
- **Global Error:** If we simulate for a fixed time τ using $L = \tau/\epsilon$ steps, the errors accumulate.

$$\text{Global Error} \approx L \times \text{Local Error} \approx \frac{1}{\epsilon} \times O(\epsilon^{p+1}) = O(\epsilon^p)$$

Leapfrog is a **2nd Order** integrator. This means one step has error ϵ^3, and the total path has error ϵ^2.
Background: Multivariate Taylor Expansion For a vector function $f(x)$, the expansion around x is:

$$f(x+\delta) = f(x) + \nabla f(x)^T \delta + \frac{1}{2!}\delta^T \mathbf{H}_f(x)\delta + \dots$$

Or for time evolution $w(t+\epsilon)$:

$$w(t+\epsilon) = w(t) + \epsilon \dot{w}(t) + \frac{\epsilon^2}{2}\ddot{w}(t) + \frac{\epsilon^3}{6}\dddot{w}(t) + \dots$$

Proof Steps
Goal: Compare the *Exact* Taylor expansion with the *Leapfrog* expansion to find the mismatch.
1. The Exact Physics (Ground Truth)
From Hamilton's equations, we know $\dot{w} = v$ and $\dot{v} = -\nabla H(w) = -g(w)$. Differentiating again, $\ddot{w} = \dot{v} = -g(w)$. So the exact position at $t+\epsilon$ is:

$$w_{exact}(t+\epsilon) = w + \epsilon v - \frac{\epsilon^2}{2}g(w) + O(\epsilon^3)$$

2. The Leapfrog Approximation
Substitute the Leapfrog equations into each other to get a single expression for w^*. From the Drift step:

$$w^* = w + \epsilon v_{1/2}$$

Substitute $v_{1/2}$ from the first Half-Kick:

$$w^* = w + \epsilon\left(v - \frac{\epsilon}{2}g(w)\right)$$

Expand this:

$$w^* = w + \epsilon v - \frac{\epsilon^2}{2}g(w)$$

3. Comparison

1. Compare the equation for w_{exact} (Step 1) and w^* (Step 2). Do they match up to the ϵ^2 term?
2. Now, perform the same expansion for the momentum v^*. *Hint: You will need to Taylor expand $g(w^*) = g(w + \Delta w)$ to see the error term.*
3. Identify the first term where they **don't** match. It should be an ϵ^3 term.

4. Global Error Argument
Since the local error (one step) is $O(\epsilon^3)$, explain why simulating a trajectory of length $\tau = 1$ requires $1/\epsilon$ steps, and therefore results in a total error of $O(\epsilon^2)$.